

2020 Trial Higher School Certificate PRELIMINARY EXAMINATION

Mathematics Standard

- Solutions including marking scale
- Mapping grid

We have endeavoured to ensure that the solutions are free of errors and follow the spirit of the syllabus in the methods used to solve the problems. However, individual teachers may opt for alternate solutions and/or may select a different marking system.

Marking Guidelines: Mathematics Standard

1B 2D 3C 4A 5D 6D 7C 8A 9D 10D 11B 12C 13C 14A 15C

ANSWERS QUESTION 16

Question 16 (a)

Criteria	Marks
Provides correct solution	1

$$4 \times \$8 + 1.5 \times \$8.60 + 5.5 \times \$2.50 = \$32 + \$12.90 + \$13.75 = \$58.65$$

Question 16 (b)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$\text{Earnings at Poles} = 2 \times 1.5 \times 7.5 + 8 \times 2 \times 7.5 = \$22.50 + \$120 = \$142.50$$

$$\text{Total earnings} = \$142.50 + \$58.65 = \$201.15$$

Question 17

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$v = \frac{4}{3}\pi(8)^3/2 = 1072.33\text{cm}^3$$

Question 18 (a)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$A = 20/2(15+18) = 330 \text{ cm}^2$$

Question 18 (b)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$A = 0.5(12)(41) + 0.5(15)(9) + 0.5(12)(18) + 330 = 751.5\text{m}^2$$

Question 19 (a)

Criteria	Marks
Provides correct solution	3
Calculates both John's and Tracey's tax paid	2
Provides some relevant information	1

$$\text{John's taxable income} = \$56\,000 - \$15\,000 = \$41\,000$$

$$\text{Tax paid on } \$41\,000 = \$2380 + 0.3(\$41\,000 - \$20\,000) = \$8680$$

$$\text{Tracey's taxable income} = \$82\,000 - \$17\,000 = \$65\,000$$

$$\text{Tax paid on } \$65\,000 = \$15\,580 + 0.47(\$65\,000 - \$60\,000) = \$17\,930$$

$$\text{Difference } \$17\,930 - \$8\,680 = \$9\,250$$

Question 19 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Total tax deductions = $\$334 \times 52 = \17368

Difference = $\$17\,930 - \$17\,368 = \$562$ (Will have to pay extra tax)

Question 20 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Stamp duty = $\$1350 + 5[(68\,000 - 45\,000)/100] = \$2\,500$

Question 20 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Total cost = $\$675 + 480 + 2500 + 1260 + 68\,000 = \$72\,915$

Question 21

Criteria	Marks
• Provides correct solution	3
• Majority of charges are given	2
• Provides some relevant information	1

Total amount = $\$0.6618 \times 91 + \$0.2475 \times 1685 + (2352 - 1685) \times 0.2781$
 $= \$60.224 + 417.0375 + 185.493 = \662.75

Question 22

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$6(2 - 3x) = -60$$

$$2 - 3x = -10$$

$$-3x = -12$$

$$x = 4$$

Question 23 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

	S	R	F
B	BS	BR	BF
G	GS	GR	GF
H	HS	HR	HF
C	CS	CR	CF

Question 23 (b)

Criteria	Marks
• Provides correct solution	1

12

Question 23 (c)

Criteria	Marks
• Provides correct solution	1

1/12

Question 24

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$S = V_0 - D_n$
 $4000 = 8\,000 - 680n$
 $680n = 4000$
 $n = 5.88$ years

Question 25 (a)

Criteria	Marks
• Provides correct solution	4
• Finds curved surface correctly and one concentric circle	3
• Provides area of one concentric circle	2
• Provides some relevant information	1

Area of concentric circle = $\pi(6)^2 - \pi(3)^2 = 84.82 = 84.8\text{m}^2$

Area = Area of two ends plus curved surface (both inside and outside)
 $= 2 \times 84.8 + 2\pi(6)(18) + 2\pi(3)(18) = 1187.47 = 1187\text{m}^2$

Question 25 (b)

Criteria	Marks
Provides correct solution	1

Number of cans = $1187/16 = 74.18$

Must buy 4 x 20 litre cans

Cost is $4 \times \$134 = \536

Question 26 (a)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$C = yA/(y + 12)$$

$$Cy + 12C = yA$$

$$A = (Cy + 12C)/y = C(y + 12)/y$$

Question 26 (b)

Criteria	Marks
Provides correct solution	1

$$A = 12(4 + 12)/4 = 48 \text{ mL}$$

Question 27 (a)

Criteria	Marks
Provides correct solution	1

9am Melbourne time = 1 pm Friday Hawaii time

Question 27 (b)

Criteria	Marks
Provides correct solution	3
Majority of the solution is correct	2
Provides some relevant information	1

10 pm (Sunday Vancouver time) → 4 pm (Monday Melbourne time). Note 18 hours difference.

The plane takes $(10 + 8 + 2) = 20$ hrs to fly from Melbourne to Vancouver.

The plane left Melbourne 8 pm Sunday (Melbourne time)

Question 28 (a)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$\text{Paula (BAC)} = (10 \times 8 - 7.5 \times 7) / (5.5 \times 63) = 0.079$$

$$\text{Tim (BAC)} = (10 \times 11 - 7.5 \times 7) / (6.8 \times 78) = 0.108$$

Question 28 (b)

Criteria	Marks
• Provides correct solution	1

$$N = 0.108/0.015 = 7.2 \text{ hours} = 7 \text{ hours and } 12 \text{ minutes}$$

Question 29 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$P = \frac{1}{2}[2\pi(3)] + 15 + 18 + \sqrt{3^2 + 6^2} = 3\pi + 33 + \sqrt{45} = 49.13 \text{ m}$$

Question 29 (b)

Criteria	Marks
• Provides correct solution	1

$$\text{Cost} = \$49.13 \times 89.65 = \$4404.50$$

Question 30

Criteria	Marks
• Provides correct solution	3
• Changes the subject to v correctly	2
• Provides some relevant information	1

$$d = 0.00565v^2$$

$$v^2 = 106.4/0.00565$$

$$v = \sqrt{\frac{106.4}{0.00565}} = 137.22 = 137 \text{ km/h}$$

Question 31 (a)

Criteria	Marks
• Provides correct solution	1

$$\text{Mean time} = 24.05 \text{ mins}$$

Question 31 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\text{Lower extreme} = 4$$

$$\text{Lower quartile} = 18$$

$$\text{Median} = 24$$

$$\text{Upper quartile} = 26 \quad \text{Upper extreme} = 44$$

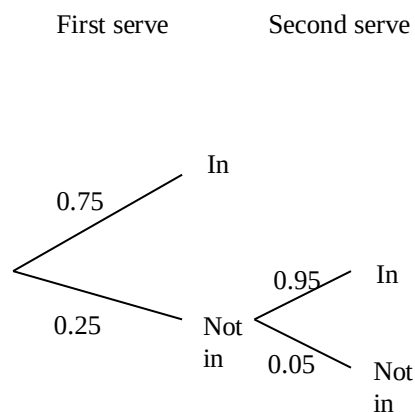
Question 31 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1



Question 32 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1



Question 32 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Prob (at least one serve will go in) = $1 - \text{Prob (both out)} = 1 - (0.25)(0.05) = 0.9875 = 98.75\%$

Question 32 (c)

Criteria	Marks
• Provides correct solution	1

Prob (double fault) = $(0.25)(0.05) = 0.0125 = 1.25\%$

Question 33 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\begin{aligned}
 v &= \frac{1}{2} \pi \times r^2 \times h \\
 &= \frac{1}{2} \times \pi \times 13.5^2 \times 12 \\
 &= 3435\text{m}^3
 \end{aligned}$$

Question 33 (b)

Criteria	Marks
• Provides correct solution	1

$$\text{Capacity} = 3435 \text{ kL}$$

Question 33 (c)

Criteria	Marks
• Provides correct solution	2
•	1

$$\begin{aligned}
 SA &= \frac{1}{2} \times 2 \times \pi \times 13.5 \times 12 + \pi(13.5)^2 \\
 &= 1081.5 \text{ square metres}
 \end{aligned}$$

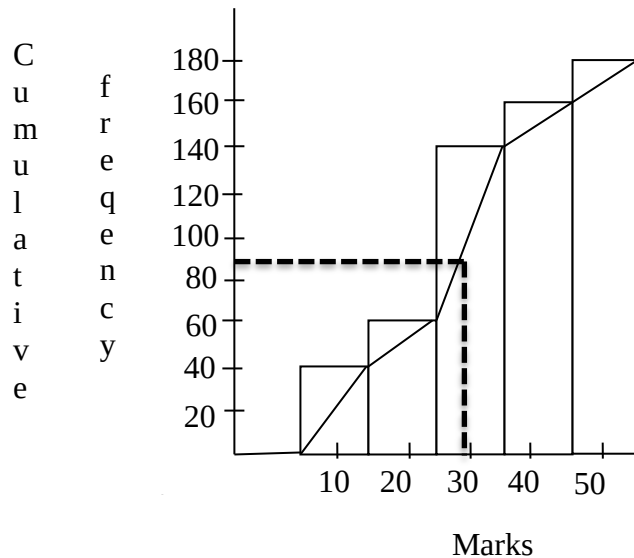
Question 34 (a)

Criteria	Marks
• Provides correct solution	1

Mark	frequency	Cumulative Frequency
10	40	40
20	20	60
30	80	140
40	20	160
50	20	180

Question 34 (b)

Criteria	Marks
• Provides correct solution	3
• Correct histogram or polygon	2
• Correct labelling of axes	1



Question 34 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

From the graph the median is 29

Question 35 (a)

Criteria	Marks
• Provides correct solution	1

Constant of variation = $80/50 = 8/5$

Question 35 (b)

Criteria	Marks
• Provides correct solution	1

$k = 8/5 m$

Question 35 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

100 km = $\frac{8}{5}$ miles

i.e. answer is 62.5 miles

Question 36 (a)

Criteria	Marks
• Provides correct solution	1

Commission = 8% of \$180 000 + 7 % of \$110 000
= \$22 100

Question 36 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Total commission = \$22 100+ 8% of \$180 000+7% of \$120000 = \$44 900

Question 36 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Pay = \$560 + 9% of \$44 900=\$4 601

Question 37

Criteria	Marks
• Provides correct solution	3
• Finds number of litres for \$46.60	2
• Provides some relevant information	1

$$13L \rightarrow 166\text{km}$$

$$1L \rightarrow \frac{166}{13} \text{ km}$$

cost to travel 166 km

$$13 \times 0.96 = \$12.48$$

$$\text{for \$1 you travel } \frac{166}{12.48} \text{ km}$$

$$\text{for \$46.60 you travel } \frac{166}{12.48} \times \$46.60 = 619.8397 \text{ km}$$

Question 38

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$S = \frac{n}{2} [2a + (n-1)d]$$

$$\frac{2S}{n} = 2a + (n-1)d$$

$$\frac{2S}{n} - 2a = (n-1)d$$

$$\frac{2S - 2an}{n} = (n-1)d$$

$$d = \frac{2S - 2an}{n} \div (n-1)$$

$$d = \frac{2S - 2an}{n(n-1)}$$

Mathematics Standard

Preliminary HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes	Targeted Performance bands
1	1	M1.1: Measurement and energy	MS11-3,4,9,10	1-2
2	1	S1.1: Classifying and representing data	MS11-2,7,8,9,10	2-3
3	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	2-3
4	1	M1.1: Measurement and energy	MS11-3,4,9,10	2-3
5	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	2-3
6	1	M 1.1: Measurement and energy	MS11-3,4,9,10	2-3
7	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	2-3
8	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
9	1	S1.1: Classifying and representing data	MS11-2,7,8,9,10	2-3
10	1	MS-A2: Linear relationships	MS11-1,6,9,10	3-4
11	1	MS-M2: Working with time	MG11-1,6,9,10	3-4
12	1	MS-A2: Linear relationships	MS11-1,6,9,10	3-4
13	1	MS-S2: Relative frequency and probability	MS11-2,7,8,9,10	3-4
14	1	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
15	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	2-3
16 (a)	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
16 (b)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
17	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
18 (a)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	2-3
18 (b)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
19 (a)	3	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
19 (b)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
20 (a)	2	F1.3: Budgeting and household expenses	MS11-2,5,6,9,10	2-3
20 (b)	2	F1.3: Budgeting and household expenses	MS11-2,5,6,9,10	2-3
21	3	M1.1: Measurement and energy	MS11-3,4,9,10	3-4
22	2	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
23 (a)	2	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
23 (b)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4

23 (c)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
24	2	F1.1: Interest and depreciation	MS11-2,5,6,9,10	3-4
25 (a)	4	M1.2: Perimeter, area and volume	MS11-3,4,9,10	4-5
25 (b)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	4-5
26 (a)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
26 (b)	1	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
27 (a)	1	MS-M2: Working with time	MG11-1,6,9,10	3-4
27 (b)	3	MS-M2: Working with time	MG11-1,6,9,10	3-4
28 (a)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
28 (b)	1	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
29 (a)	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
29 (b)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	2-3
30	3	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
31 (a)	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	3-4
31 (b)	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	3-4
31 (c)	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	3-4
32 (a)	2	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
32 (b)	2	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
32 (c)	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
33 (a)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	5-6
33 (b)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	5-6
33 (c)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	5-6
34 (a)	1	S1.1: Classifying and representing data	MS11-2,7,8,9,10	3-4
34 (b)	3	S1.1: Classifying and representing data	MS11-2,7,8,9,10	3-4
34 (c)	2	S1.1: Classifying and representing data	MS11-2,7,8,9,10	3-4
35 (a)	1	MS-A2: Linear relationships	MS11-1,6,9,10	3-4
35 (b)	1	MS-A2: Linear relationships	MS11-1,6,9,10	3-4
35 (c)	2	MS-A2: Linear relationships	MS11-1,6,9,10	3-4
36 (a)	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
36 (b)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
36 (c)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
37	3	F1.3: Budgeting and household expenses	MS11-2,5,6,9,10	2-3
38	2	MS-A1: Formulae and equations	MS11-1,6,9,10	4-5